HW5

RJ Cass

```
suppressPackageStartupMessages(library(dplyr))
suppressPackageStartupMessages(library(lme4))
suppressPackageStartupMessages(library(multcomp))
suppressPackageStartupMessages(library(emmeans))
cult <- read.table('cult_data.dat', header = FALSE)
colnames(cult) <- c('field', 'cultivar', 'soil', 'weight')
cult <- cult %>%
    group_by(soil, cultivar) %>%
    mutate(group = cur_group_id()) %>%
    ungroup()
cult <- cult %>%
    mutate(across(c(field, cultivar, soil, group), as.factor))
```

1

```
fit <- lmer(weight ~  cultivar * soil + (1|field), data=cult)
summary(fit)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: weight ~ cultivar * soil + (1 | field)
   Data: cult

REML criterion at convergence: 68.5

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.40580 -0.53790  0.06846  0.59451  1.84385

Random effects:
```

```
 Groups   Name        Variance Std.Dev.
 field    (Intercept) 1.207    1.099
 Residual             1.196    1.094
Number of obs: 24, groups:  field, 4

Fixed effects:
                 Estimate Std. Error t value
(Intercept)       27.9000     0.7752  35.992
cultivarb          0.1250     0.7734   0.162
soildea            1.3500     0.7734   1.746
soilliv            5.2500     0.7734   6.788
cultivarb:soildea  1.2750     1.0938   1.166
cultivarb:soilliv  0.2500     1.0938   0.229

Correlation of Fixed Effects:
            (Intr) cltvrb soilde soillv cltvrb:sld
cultivarb   -0.499
soildea     -0.499  0.500
soilliv     -0.499  0.500  0.500
cultvrb:sld  0.353 -0.707 -0.707 -0.354
cltvrb:sllv  0.353 -0.707 -0.354 -0.707  0.500
```

The variance component of the random error is the 'Residual Variance' in the 'Random Effects' portion: 1.196

2

The between field variance component is the 'Field Variance' in the 'Random Effects' portion: 1.207

3

```r
cell_means_mod <- lmer(weight ~ group - 1 + (1|field), data = cult)
summary(cell_means_mod)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: weight ~ group - 1 + (1 | field)
   Data: cult
```

```
REML criterion at convergence: 68.5

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.40580 -0.53790  0.06846  0.59451  1.84385

Random effects:
 Groups   Name        Variance Std.Dev.
 field    (Intercept) 1.207    1.099
 Residual             1.196    1.094
Number of obs: 24, groups:  field, 4

Fixed effects:
       Estimate Std. Error t value
group1  27.9000     0.7752   35.99
group2  28.0250     0.7752   36.15
group3  29.2500     0.7752   37.73
group4  30.6500     0.7752   39.54
group5  33.1500     0.7752   42.76
group6  33.5250     0.7752   43.25

Correlation of Fixed Effects:
       group1 group2 group3 group4 group5
group2 0.502
group3 0.502  0.502
group4 0.502  0.502  0.502
group5 0.502  0.502  0.502  0.502
group6 0.502  0.502  0.502  0.502  0.502
```

The estimate of the cell means is the estimates from the 'Fixed Effects' portion:

Group	Estimate
1	27.9
2	28.025
3	29.25
4	30.65
5	33.15
6	33.525

4

The dimension of $\hat{\Sigma}$ is $n \times n$ (24×24)

```
sigma_e <- 1.163
sigma_b <- 1.213

Z <- model.matrix(weight ~ field - 1, data = cult)
G <- sigma_b * diag(length(unique(cult$field)))
R <- sigma_e * diag(nrow(cult))

sigma_hat <- Z %*% G %*% t(Z) + R
sigma_hat
```

```
       1     2     3     4     5     6     7     8     9    10    11    12
1  2.376 1.213 1.213 1.213 1.213 1.213 0.000 0.000 0.000 0.000 0.000 0.000
2  1.213 2.376 1.213 1.213 1.213 1.213 0.000 0.000 0.000 0.000 0.000 0.000
3  1.213 1.213 2.376 1.213 1.213 1.213 0.000 0.000 0.000 0.000 0.000 0.000
4  1.213 1.213 1.213 2.376 1.213 1.213 0.000 0.000 0.000 0.000 0.000 0.000
5  1.213 1.213 1.213 1.213 2.376 1.213 0.000 0.000 0.000 0.000 0.000 0.000
6  1.213 1.213 1.213 1.213 1.213 2.376 0.000 0.000 0.000 0.000 0.000 0.000
7  0.000 0.000 0.000 0.000 0.000 0.000 2.376 1.213 1.213 1.213 1.213 1.213
8  0.000 0.000 0.000 0.000 0.000 0.000 1.213 2.376 1.213 1.213 1.213 1.213
9  0.000 0.000 0.000 0.000 0.000 0.000 1.213 1.213 2.376 1.213 1.213 1.213
10 0.000 0.000 0.000 0.000 0.000 0.000 1.213 1.213 1.213 2.376 1.213 1.213
11 0.000 0.000 0.000 0.000 0.000 0.000 1.213 1.213 1.213 1.213 2.376 1.213
12 0.000 0.000 0.000 0.000 0.000 0.000 1.213 1.213 1.213 1.213 1.213 2.376
13 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
14 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
15 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
16 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
17 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
18 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
19 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
20 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
21 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
22 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
23 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
24 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
      13    14    15    16    17    18    19    20    21    22    23    24
1  0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
2  0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
```

```
3  0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
4  0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
5  0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
6  0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
7  0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
8  0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
9  0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
10 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
11 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
12 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000
13 2.376 1.213 1.213 1.213 1.213 1.213 0.000 0.000 0.000 0.000 0.000 0.000
14 1.213 2.376 1.213 1.213 1.213 1.213 0.000 0.000 0.000 0.000 0.000 0.000
15 1.213 1.213 2.376 1.213 1.213 1.213 0.000 0.000 0.000 0.000 0.000 0.000
16 1.213 1.213 1.213 2.376 1.213 1.213 0.000 0.000 0.000 0.000 0.000 0.000
17 1.213 1.213 1.213 1.213 2.376 1.213 0.000 0.000 0.000 0.000 0.000 0.000
18 1.213 1.213 1.213 1.213 1.213 2.376 0.000 0.000 0.000 0.000 0.000 0.000
19 0.000 0.000 0.000 0.000 0.000 0.000 2.376 1.213 1.213 1.213 1.213 1.213
20 0.000 0.000 0.000 0.000 0.000 0.000 1.213 2.376 1.213 1.213 1.213 1.213
21 0.000 0.000 0.000 0.000 0.000 0.000 1.213 1.213 2.376 1.213 1.213 1.213
22 0.000 0.000 0.000 0.000 0.000 0.000 1.213 1.213 1.213 2.376 1.213 1.213
23 0.000 0.000 0.000 0.000 0.000 0.000 1.213 1.213 1.213 1.213 2.376 1.213
24 0.000 0.000 0.000 0.000 0.000 0.000 1.213 1.213 1.213 1.213 1.213 2.376
```

5

```r
LambdaT <- getME(cell_means_mod, "Lambdat")
Zt      <- getME(cell_means_mod, "Zt")

# Extract the residual variance scale (sigma^2)
vr <- sigma(cell_means_mod)^2

# Combine them to get the total covariance matrix (V)
var_b <- vr * (t(Zt) %*% crossprod(LambdaT) %*% Zt)
sI    <- vr * Matrix::Diagonal(nobs(cell_means_mod))
V     <- var_b + sI

# View V as a standard dense matrix
as.matrix(V)
```

```
          1        2        3        4        5        6        7        8
```

1	2.403611	1.207278	1.207278	1.207278	1.207278	1.207278	0.000000	0.000000
2	1.207278	2.403611	1.207278	1.207278	1.207278	1.207278	0.000000	0.000000
3	1.207278	1.207278	2.403611	1.207278	1.207278	1.207278	0.000000	0.000000
4	1.207278	1.207278	1.207278	2.403611	1.207278	1.207278	0.000000	0.000000
5	1.207278	1.207278	1.207278	1.207278	2.403611	1.207278	0.000000	0.000000
6	1.207278	1.207278	1.207278	1.207278	1.207278	2.403611	0.000000	0.000000
7	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	2.403611	1.207278
8	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.207278	2.403611
9	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.207278	1.207278
10	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.207278	1.207278
11	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.207278	1.207278
12	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	1.207278	1.207278
13	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
14	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
15	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
16	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
17	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
18	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
19	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
20	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
21	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
22	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
23	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
24	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000

	9	10	11	12	13	14	15	16
1	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
2	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
3	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
5	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
6	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
7	1.207278	1.207278	1.207278	1.207278	0.000000	0.000000	0.000000	0.000000
8	1.207278	1.207278	1.207278	1.207278	0.000000	0.000000	0.000000	0.000000
9	2.403611	1.207278	1.207278	1.207278	0.000000	0.000000	0.000000	0.000000
10	1.207278	2.403611	1.207278	1.207278	0.000000	0.000000	0.000000	0.000000
11	1.207278	1.207278	2.403611	1.207278	0.000000	0.000000	0.000000	0.000000
12	1.207278	1.207278	1.207278	2.403611	0.000000	0.000000	0.000000	0.000000
13	0.000000	0.000000	0.000000	0.000000	2.403611	1.207278	1.207278	1.207278
14	0.000000	0.000000	0.000000	0.000000	1.207278	2.403611	1.207278	1.207278
15	0.000000	0.000000	0.000000	0.000000	1.207278	1.207278	2.403611	1.207278
16	0.000000	0.000000	0.000000	0.000000	1.207278	1.207278	1.207278	2.403611
17	0.000000	0.000000	0.000000	0.000000	1.207278	1.207278	1.207278	1.207278
18	0.000000	0.000000	0.000000	0.000000	1.207278	1.207278	1.207278	1.207278

```
19 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
20 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
21 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
22 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
23 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
24 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
         17       18       19       20       21       22       23       24
1  0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
2  0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
3  0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
4  0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
5  0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
6  0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
7  0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
8  0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
9  0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
10 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
11 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
12 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
13 1.207278 1.207278 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
14 1.207278 1.207278 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
15 1.207278 1.207278 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
16 1.207278 1.207278 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
17 2.403611 1.207278 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
18 1.207278 2.403611 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000
19 0.000000 0.000000 2.403611 1.207278 1.207278 1.207278 1.207278 1.207278
20 0.000000 0.000000 1.207278 2.403611 1.207278 1.207278 1.207278 1.207278
21 0.000000 0.000000 1.207278 1.207278 2.403611 1.207278 1.207278 1.207278
22 0.000000 0.000000 1.207278 1.207278 1.207278 2.403611 1.207278 1.207278
23 0.000000 0.000000 1.207278 1.207278 1.207278 1.207278 2.403611 1.207278
24 0.000000 0.000000 1.207278 1.207278 1.207278 1.207278 1.207278 2.403611
```

6

```r
# Get the contrast matrix to apply directly to the model (coefs pulled from fixef(fit))
L <- matrix(c(0, 1, 0, 0, 0, 0), nrow = 1)
colnames(L) <- names(fixef(fit))
cultivar_test <- glht(fit, linfct = L)
summary(cultivar_test)
```

```
     Simultaneous Tests for General Linear Hypotheses

Fit: lmer(formula = weight ~ cultivar * soil + (1 | field), data = cult)

Linear Hypotheses:
       Estimate Std. Error z value Pr(>|z|)
1 == 0   0.1250     0.7734   0.162    0.872
(Adjusted p values reported -- single-step method)
```

No, the effect is not significant.

7

```r
L <- matrix(c(
  0, 0, 1, 0, 0, 0, # Row 1: Tests if soil2 = 0
  0, 0, 0, 1, 0, 0 # Row 2: Tests if soil3 = 0
), nrow = 2, byrow = TRUE)
colnames(L) <- names(fixef(fit))
soil_joint_test <- glht(fit, linfct = L)
summary(soil_joint_test, test = Chisqtest())
```

```
     General Linear Hypotheses

Linear Hypotheses:
       Estimate
1 == 0     1.35
2 == 0     5.25

Global Test:
  Chisq DF Pr(>Chisq)
1  49.7  2  1.612e-11
```

Yes, the effect is significant.

8

```
L <- matrix(c(
  0, 0, 0, 0, 1, 0,  # Row 1: Tests if cultivarB:soil2 = 0
  0, 0, 0, 0, 0, 1   # Row 2: Tests if cultivarB:soil3 = 0
), nrow = 2, byrow = TRUE)
colnames(L) <- names(fixef(fit))
interaction_test <- glht(fit, linfct = L)
summary(interaction_test, test = Chisqtest())
```

```
     General Linear Hypotheses

Linear Hypotheses:
       Estimate
1 == 0    1.275
2 == 0    0.250

Global Test:
  Chisq DF Pr(>Chisq)
1 1.526  2     0.4662
```

No, there are not significant interactions.

9

```
cult_red <- cult[!(cult$cultivar == 'b' & cult$soil == 'liv'),]
fit_red <- lmer(weight ~ cultivar * soil + (1|field), data = cult_red)
```

```
fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

```
summary(fit_red)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: weight ~ cultivar * soil + (1 | field)
   Data: cult_red
```

```
REML criterion at convergence: 53.4

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.3548 -0.4318 -0.1165  0.6612  1.2888

Random effects:
 Groups   Name        Variance Std.Dev.
 field    (Intercept) 0.7525   0.8675
 Residual             0.9407   0.9699
Number of obs: 20, groups:  field, 4

Fixed effects:
                  Estimate Std. Error t value
(Intercept)        27.9000     0.6506  42.883
cultivarb           0.1250     0.6858   0.182
soildea             1.3500     0.6858   1.968
soilliv             5.2500     0.6858   7.655
cultivarb:soildea   1.2750     0.9699   1.315

Correlation of Fixed Effects:
            (Intr) cltvrb soilde soillv
cultivarb   -0.527
soildea     -0.527  0.500
soilliv     -0.527  0.500  0.500
cultvrb:sld  0.373 -0.707 -0.707 -0.354
fit warnings:
fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

Random Error: .9407

Between Field Variance: .7525

10

```
cult_red$group <- droplevels(cult_red$group)
cell_means_mod_red <- lmer(weight ~ cultivar * soil + (1|field), data = cult_red)
```

```
fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

```r
summary(cell_means_mod_red)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: weight ~ cultivar * soil + (1 | field)
   Data: cult_red

REML criterion at convergence: 53.4

Scaled residuals:
    Min      1Q  Median      3Q     Max
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Number of obs: 20, groups:  field, 4

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cultivarb           0.1250     0.6858   0.182
soildea             1.3500     0.6858   1.968
soilliv             5.2500     0.6858   7.655
cultivarb:soildea   1.2750     0.9699   1.315

Correlation of Fixed Effects:
            (Intr) cltvrb soilde soillv
cultivarb   -0.527
soildea     -0.527  0.500
soilliv     -0.527  0.500  0.500
cultvrb:sld  0.373 -0.707 -0.707 -0.354
fit warnings:
fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

Group	Estimate
1	27.9
2	28.025
3	29.25
4	30.65

Group	Estimate
5	33.15

11

```
cell_means_red_filled <- emmeans(fit_red, ~ cultivar * soil)
print(cell_means_red_filled)
```

```
 cultivar soil emmean    SE   df lower.CL upper.CL
 a        con    27.9 0.651 8.38     26.4     29.4
 b        con    28.0 0.651 8.38     26.5     29.5
 a        dea    29.2 0.651 8.38     27.8     30.7
 b        dea    30.6 0.651 8.38     29.2     32.1
 a        liv    33.1 0.651 8.38     31.7     34.6
 b        liv  nonEst    NA   NA       NA       NA

Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95
```

```
summary(fit_red)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: weight ~ cultivar * soil + (1 | field)
   Data: cult_red

REML criterion at convergence: 53.4

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.3548 -0.4318 -0.1165  0.6612  1.2888

Random effects:
 Groups   Name        Variance Std.Dev.
 field    (Intercept) 0.7525   0.8675
 Residual             0.9407   0.9699
Number of obs: 20, groups:  field, 4

Fixed effects:
```

```
                  Estimate Std. Error t value
(Intercept)        27.9000     0.6506  42.883
cultivarb           0.1250     0.6858   0.182
soildea             1.3500     0.6858   1.968
soilliv             5.2500     0.6858   7.655
cultivarb:soildea   1.2750     0.9699   1.315

Correlation of Fixed Effects:
            (Intr) cltvrb soilde soillv
cultivarb   -0.527
soildea     -0.527  0.500
soilliv     -0.527  0.500  0.500
cultvrb:sld  0.373 -0.707 -0.707 -0.354
fit warnings:
fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

```r
C <- matrix(c(
  # Intercept, Cultivar_B, SoilDea, SoilLiv, CultB:SoilDea, CultB:SoilLiv
  1, 0, 0, 0, 0, 0,  # Group 1 structure
  1, 1, 0, 0, 0, 0,  # Group 2 structure
  1, 0, 1, 0, 0, 0,  # Group 3 structure
  1, 1, 1, 0, 1, 0,  # Group 4 structure
  1, 0, 0, 1, 0, 0,  # Group 5 structure
  1, 1, 0, 1, 0, 1
), nrow = 6, byrow = TRUE)

fit_red_beta <- lm(weight ~ cultivar * soil, data = cult_red)
betas <- coef(fit_red_beta)
betas[is.na(betas)] <- 0

# 4. Multiply your C matrix by R's actual coefficients
mu_check <- C %*% betas
print(mu_check)
```

```
       [,1]
[1,] 27.900
[2,] 28.025
[3,] 29.250
[4,] 30.650
[5,] 33.150
[6,] 33.275
```

The estimate for the missing cell is 33.9. When all data is inclued the estimate is 33.525, so this slightly overestimated the mean. However, it is within one standard error so pretty close!

12

```
vcov_raw <- vcov(fit_red_beta)

vcov_raw <- vcov(fit_red_beta)

vcov_full <- vcov_raw
vcov_full[, 6] <- 0
vcov_full[6, ] <- 0

var_cell_means <- C %*% vcov_full %*% t(C)

group_variances <- diag(var_cell_means)

group_standard_errors <- sqrt(group_variances)

# ---- Display the Results ----
results_table <- data.frame(
  Group = 1:6,
  Estimate = mu_check,
  Variance = group_variances,
  Std_Error = group_standard_errors
)
print(results_table)
```

```
  Group Estimate  Variance Std_Error
1     1   27.900 0.4232917 0.6506087
2     2   28.025 0.4232917 0.6506087
3     3   29.250 0.4232917 0.6506087
4     4   30.650 0.4232917 0.6506087
5     5   33.150 0.4232917 0.6506087
6     6   33.275 1.2698750 1.1268873
```

The Error for group 6 is much larger than that of the other groups, which is expected since we are imputing data that "doesn't" exist.